

605.52 g 614.66 g
604.85 g 614.17 g

HighSTEPS Experiment

Exp. Name: h380
Operator: Barnaby

Date: 26-9-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G2658 Fresh Tichy andesite
☐ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.02
Horz. Load Initial Value (kN): -0.41

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/sec for 11mm

Acceleration: 566 μ m/s²
Deceleration: _____

Starting Position: _____
Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): RH

Comments:

Thinnest spacer used for setup
Held 10 MPa 14:04 - 14:24

606.15g 605.37g
606.53g 615.43g

HighSTEPS Experiment

Exp. Name: h381
Operator: Barnaby

Date: 27-9-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G2658 Fresh Trachyandesite
☐ Gouge - Particle Size, Size Distribution: 425 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0.0
Horz. Load Initial Value (kN): -0.32

Horizontal Load PID Setting:
gain: 0.003
Ti: 0.52
Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/sec for 11 mm

Acceleration: 5e6 μ m/s²
Deceleration: _____

Starting Position: _____
Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): wet

Comments:

Thinnest square used for setup
Saturated at 1MPa 9:37 - 10:17

605.54g 614.98g
604.49g 613.97g

HighSTEPS Experiment

Exp. Name: W382
Operator: Barnaby

Date: 27-9-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G2658 Fresh Trachyandesite
☐ Gouge - Particle Size, Size Distribution: < 125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.1
Horz. Load Initial Value (kN): -0.44

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 34.68 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/sec for 11mm

Acceleration: 5e6 μ m/s²
Deceleration: _____

Starting Position: _____
Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): RH

Comments:

thinnest square used for setup

606.12g 615.28g
608.57g 615.70g

HighSTEPS Experiment

Exp. Name: h383
Operator: Barnaby

Date: 28-9-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G2658 Fresh Trachyandesite
☐ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.16
Horz. Load Initial Value (kN): -0.38

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 34.68 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/sec for 11mm

Acceleration: 5e6 μ m/s²
Deceleration: _____

Starting Position: _____
Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): wet

Comments:

thinnest spacer used for setup
saturated @ 1MPa 7:38 - 8:18
held 30 MPa 9:14 - 9:34

606.12g 615.14g
606.54g 615.76g

HighSTEPS Experiment

Exp. Name: 4421
Operator: Barnaby

Date: 26-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G-2658

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.2
Horz. Load Initial Value (kN): -0.65

Horizontal Load PID Setting:

gain: 0.033
Ti: 0.02
Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 5e6 μ m/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-5-10-30-100-200
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): dry

Comments:

thinnest square used for setup

605.54g 614.93g
604.85g 614.44g

HighSTEPS Experiment

Exp. Name: h422
Operator: Barnaby

Date: 27-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G-2658

☒ Gouge - Particle Size, Size Distribution: <125µm
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.1
Horz. Load Initial Value (kN): -0.42

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10µm/s for 11mm

Acceleration: 5e6µm/s²
Deceleration: _____

Starting Position: _____
Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): as wet

Comments:

thinnest square used for setup
Saturated @ 1MPa 9:03-9:43

606.15g 013.87g
606.60g 615.18g

HighSTEPS Experiment

Exp. Name: h385
Operator: Barnaby

Date: 29-9-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G2690 Pink altered
☐ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.12
Horz. Load Initial Value (kN): -0.35

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 5e6 μ m/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): dry

Comments:

Thinnest square used for setup

606.18g 614.76g
606.59g 615.20g

HighSTEPS Experiment

Exp. Name: h386
Operator: Barnaby

Date: 30-9-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G2690 Pink altered
☐ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0.0
Horz. Load Initial Value (kN): -0.3

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/sec for 11mm

Acceleration: 5e6 μ m/s²
Deceleration: _____

Starting Position: 283.7, 39.2
Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300
Holds: 1-3-10-30-400-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): wet

Comments:

thinnest square used for setup
Saturated @ 1MPa 9:00 - 9:40

605.50 g 614.20 g
604.90 g 613.54 g

HighSTEPS Experiment

Exp. Name: h387
Operator: Barnaby

Date: 30-9-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G 2690 Pink altered
☐ Gouge - Particle Size, Size Distribution: < 125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.17
Horz. Load Initial Value (kN): -0.44

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 34.68 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 5e6 μ m/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300
Holds: 1-3-10-30-100-300-1000-2000

Temperature (°C): RT

Relative Humidity (%): dry

Comments:

thinnest square used for setup

605.48g 613.93g
604.89g 613.40g

HighSTEPS Experiment

Exp. Name: h384
Operator: Barnaby

Date: 28-9-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G2690 Pink altered
☐ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.21
Horz. Load Initial Value (kN): -0.5

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 34.68 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 5e6 μ m/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-20-60-300
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): wet

Comments:

thinnest square used for setup
Saturated @ 1MPa 11:53-12:33

606.53g 615.02g II
606.13g 614.98g I

HighSTEPS Experiment

Exp. Name: h423
Operator: Baird

Date: 27-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G 2690

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.2

Horz. Load Initial Value (kN): -0.63

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11 mm

Acceleration: 526 μ m/s²

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): dry

Comments:

thinnest square used for setup

605.53g 614.36g
604.86g 613.55g

HighSTEPS Experiment

Exp. Name: h424
Operator: Barnaby

Date: 28-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G2690

☒ Gouge - Particle Size, Size Distribution: 425 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.1
Horz. Load Initial Value (kN): -0.4

Horizontal Load PID Setting:

gain: 0.005
Ti: 0.02
Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 5e6 μ m/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): wet

Comments:

thinnest square used for setup

Saturated @ 1 MPa 9:00- 9:41

One hold was ugly

606.16g 614.6g
606.61g 615.15g

HighSTEPS Experiment

Exp. Name: h393
Operator: Barnaby

Date: 7-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G3005 carbonate altered

☒ Gouge - Particle Size, Size Distribution: 2125 μ m

☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.05

Horz. Load Initial Value (kN): -0.4

Horizontal Load PID Setting:

gain: 0.03

Ti: 0.02

Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 566 μ m/s²

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): dry

Comments:

thinnest square used for setup

605.51g 613.84g
604.87g 613.44g

HighSTEPS Experiment

Exp. Name: h389
Operator: Balnaby

Date: 3-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G3005 carbonate altered

Gouge - Particle Size, Size Distribution: <125 μ m

Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0.1

Horz. Load Initial Value (kN): -0.12

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 566 μ m/s²

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10⁻³ - 10⁻³⁰ - 100 - 300

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): wet

Comments:

Thinnest square used for setup
saturated @ 1MPa 9:11 - 9:51
small leak but manageable

606.13g 614.70g
606.56g 615.35g

HighSTEPS Experiment

Exp. Name: h390
Operator: Barnaby

Date: 3-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G3005 carbonate alkred

☐ Gouge - Particle Size, Size Distribution: <125 μ m

☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0.15

Horz. Load Initial Value (kN): -0.3

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 34.68 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 5e6 μ m/s²

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300

Holds: 1-5-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): dry

Comments:

thinnest square used for setup

606.13g 614.62g
606.55g 615.04g

HighSTEPS Experiment

Exp. Name: h388
Operator: Bainaly

Date: 2-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G3005 carbonate altered
☐ Gouge - Particle Size, Size Distribution: 425 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0.1
Horz. Load Initial Value (kN): -0.25

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 34.68 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 5e6 μ m/s²
Deceleration: _____

Starting Position: _____
Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): wet

Comments:

thinnest square used for setup
Saturated @ 1 MPa 13:55-14:35
held 30 MPa 15:31-15:57
Hypiston noise @ 8200 sec
fairly clean

606.14g 614.96g
606.56g 615.34g

HighSTEPS Experiment

Exp. Name: h425
Operator: Bavinsky

Date: 28-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G3005 Carbonate altered

☒ Gouge - Particle Size, Size Distribution: 2125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.2
Horz. Load Initial Value (kN): -0.6

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 5e6 μ m/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-50-30-100-300
Holds: 1-3-10-30-100-200-1000-3000

Temperature (°C): RT
Relative Humidity (%): dry

Comments:

thinnest square used for setup

605.53g 614.55g
604.86g 613.72g

HighSTEPS Experiment

Exp. Name: h426
Operator: Barnaby

Date: 31-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G-3005 Carbonate altered

☒ Gouge - Particle Size, Size Distribution: <125µm
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0
Horz. Load Initial Value (kN): -0.2

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 µm/s for 11mm

Acceleration: 5e6 µm/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): wet

Comments:

thinnest square used for xhp
saturated @ 1MPa @ 42 - 9:23

larger } 615.10g
606.59g } 615.75g
~606

HighSTEPS Experiment

Exp. Name: H397
Operator: Bainby

Date: 10-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G330 chlorite altered

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.2
Horz. Load Initial Value (kN): -0.5

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 5e6 μ m/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300-
Holds: 1-3-10-70-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): dry

Comments:

thinnest square used for setup

606.12g 614.69g
606.55g 615.48g

HighSTEPS Experiment

Exp. Name: h395
Operator: Balinday

Date: 10-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): 63130 chlorite altered

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0.03
Horz. Load Initial Value (kN): -0.45

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 5.6 μ m/s²
Deceleration: _____

Starting Position: _____
Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): wet

Comments:

thinnest square used for setup
Saturated @ 1MPa 8:59 - 9:39
↳ small leak, manageable, sample stayed saturated
1sec hold is ugly

605.86g 614.44g
604.89g 613.73g

HighSTEPS Experiment

Exp. Name: h396
Operator: Barnaby

Date: 6-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G3130 chlorite altered
☐ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.08
Horz. Load Initial Value (kN): -0.52

Horizontal Load PID Setting:

gain: 0.03
Ti: 0.02
Td: 0.001

Normal Stress(es): 34.68 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11 mm

Acceleration: 566 μ m/s²
Deceleration: _____

Starting Position: _____
Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): dry

Comments:

thinnest square used for setup
held 30 MPa 13:18 - 13:58

605.48g 614.28g
604.88g 613.52g

HighSTEPS Experiment

Exp. Name: h394
Operator: Barnaby

Date: 8-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G3/30 chbnite altered

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0
Horz. Load Initial Value (kN): -0.35

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 34.68 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 566 μ m/s²

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-5-10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): wet

Comments:

thinnest square used for setup

Saturated @ 1 MPa 14:30-15:10

held 30 MPa 16:05-16:25

606.14g 614.77g
606.57g 615.63g

HighSTEPS Experiment

Exp. Name: h427
Operator: Burmy

Date: 31-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinner mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G-3130 chlorite altered

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.1
Horz. Load Initial Value (kN): -0.5

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 5 cm/s^2
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-30-10-30-100-300
Holds: 1-3-10-30-100-300-600-3000

Temperature ($^{\circ}\text{C}$): RT

Relative Humidity (%): dry

Comments:

thinner square used for setup

605.52g 614.23g
604.86g 613.73g

HighSTEPS Experiment

Exp. Name: h428
Operator: Barnaby

Date: 1-11-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): G3130 chlorite altered

☒ Gouge - Particle Size, Size Distribution: <125µm
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.1

Horz. Load Initial Value (kN): -0.36

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 0µm/s for 11mm

Acceleration: 506µm/s²

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-7-10-30-100-300

Holds: 1-7-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): Wet

Comments:

thinnest square used for setup

Saturated @ 1MPa 8:48 - 9:29

606.53g 614.65g
604.83g 613.28g

HighSTEPS Experiment

Exp. Name: h501
Operator: Barnaby

Date: 26-3-23

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm
Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): M1430
☒ Gouge - Particle Size, Size Distribution: 425µm
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.175
Horz. Load Initial Value (kN): 0.1

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 1.56 (kN) 10 (MPa)

Confining Pressure: / (MPa)
Pore Pressure SX: / (MPa)
Pore Pressure DX: / (MPa)

Sliding Velocity: 0µm/s 611mm

Acceleration: 56µm/s²
Deceleration: _____

Starting Position: _____
Shear Displ. Sample: _____

Velocity Steps: 10-3-10-20-100-300
Holds: 1-3-10-20-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): dry

Comments:

thinnest sphere used for setup

606.12g 614.47g
605.50g 613.96g

HighSTEPS Experiment

Exp. Name: h502
Operator: Baird

Date: 26-3-23

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): M1430

☒ Gouge - Particle Size, Size Distribution: 4125µm
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.2
Horz. Load Initial Value (kN): -0.03

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10µm/s for 11mm

Acceleration: 566µm/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3 - 10-30 - 10-300
Holds: 1-3 - 10 - 30-100 - 300 - 1000 - 3000

Temperature (°C): RT
Relative Humidity (%): wet

Comments:

thinnest square used for setup
saturated @ 1MPa 13:30 - 14:10

606.12g 614.46g
605.49g 614.15g

HighSTEPS Experiment

Exp. Name: h506
Operator: Barnaby

Date: 27-3-23

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): M1430 M4 Meager
☒ Gouge - Particle Size, Size Distribution: 425µm
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.35
Horz. Load Initial Value (kN): -0.23

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 34.68 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10µm/s for 11mm

Acceleration: 5e6µm/s²
Deceleration: _____

Starting Position: _____
Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): dry

Comments: thinnest square used for setup

606.12g 64.89g
608.47g 64.04g

HighSTEPS Experiment

Exp. Name: h510
Operator: Burnaby

Date: 29-3-23

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): M1430 Mt-Meager well M-17

☒ Gouge - Particle Size, Size Distribution: 425µm
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.45

Horz. Load Initial Value (kN): -0.16

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 34.68 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10µm/s for 11mm

Acceleration: 5e6µm/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300

Holds: 1-3-10-20-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): wet

Comments: smallest sphere used for setup
Set @ 1MPa 9:19 - 9:59

606.12g 614.67g
605.43g 613.86g

HighSTEPS Experiment

Exp. Name: h508
Operator: Burnaby

Date: 28-3-23

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): M1430 Mt Meager well M-17 1430 ft

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.41

Horz. Load Initial Value (kN): -0.27

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 526 μ m/s²

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300-1000

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): dry

Comments:

thinnest spacers used for setup

606.50g 614.94g
604.85g 613.41g

HighSTEPS Experiment

Exp. Name: h507
Operator: Barnaby

Date: 28-3-23

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): M1430 M+Meget well M17

☒ Gouge - Particle Size, Size Distribution: 425µm
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.35

Horz. Load Initial Value (kN): -0.12

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10µm/s for 11mm

Acceleration: 5e6µm/s²

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): wet

Comments:

thinnest square used for setup

Sat @ 1mb 9:18 - 9:58

604,87
605,51

611,95
614,47

HighSTEPS Experiment

Exp. Name: h365
Operator: Florian

Date: 1.9.22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N 3583

☒ Gouge - Particle Size, Size Distribution: < 125
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0.14

Horz. Load Initial Value (kN): -0.18

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 11.58 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 $\mu\text{m/s}$

Acceleration: 10^6 ($5 \cdot 10^6$) $\mu\text{m/s}^2$

Deceleration: "

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 3 - 10 - 30 - 100 - 300

Holds: 1 - 3 - 10 - 30 - 100 - 300 - 1000 - 3000

Temperature (°C): RT

Relative Humidity (%): RH

Comments:

thinnest square for sample prep

607,82 613,79
605,46 614,37

HighSTEPS Experiment

Exp. Name: h 363

Date: 31.8.22

Operator: Borisy / Florian

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
___ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
___ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
___ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
___ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N 3583 Na-Serry Sessolt N2

☒ Gouge - Particle Size, Size Distribution: <125

___ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0,13

Horz. Load Initial Value (kN): -0,35 / -0,25

Horizontal Load PID Setting:

gain: 0,007

Ti: 0,02

Td: 0,001

Normal Stress(es): 11,56 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 $\mu\text{m/s}$

Acceleration: 10⁶ (5.10⁶) $\mu\text{m/s}^2$

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 3 - 10 - 30 - 100 - 300

Holds: 1 - 3 - 10 - 30 - 100 - 300 - 1000 - 3000

Temperature (°C): RT

Relative Humidity (%): saturated

Comments:

thinnest square for sample preparation
saturation: 7h54 - 8h34

607,84 614,00
605,52 614,26

HighSTEPS Experiment

Exp. Name: h764

Date: 31.8.22

Operator: Bernady Florian

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x37 mm

☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm

☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm

☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm

☐ Porous Frits: _____ mm

Spacers: Thinest _____ mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N 3583 Newberry NZ Salt

☒ Gouge - Particle Size, Size Distribution: < 125 μ

☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0,03

Horz. Load Initial Value (kN): -0,44

Horizontal Load PID Setting:

gain: 0,007

Ti: 0,02

Td: 0,001

Normal Stress(es): ~~456~~ 34,08 (kN) to 30 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ /s

Acceleration: 1,6 (5-6 μ /s²)

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): RH

Comments:

Thinest square for sample prep

605,48 617,60
604,85 613,88

HighSTEPS Experiment

Exp. Name: h362
Operator: Bernady / Floria

Date: 30.8.22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 74x74 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N 3587 Sorelt Newbery N2

☒ Gouge - Particle Size, Size Distribution: <125 μ
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0.05

Horz. Load Initial Value (kN): -0.5

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 34.68 (kN) 70 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ /s

Acceleration: 10⁶ (5.10⁵) μ /s²

Deceleration: 5

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: ~~10-30-10~~ 10-3-10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): saturated

Comments:

thinnest square for sample prep

saturation: 12h32 - 13h17

30s hold at 15kPa

606.13g 614.72g
606.56g 615.34g

HighSTEPS Experiment

Exp. Name: h417
Operator: Balaboy

Date: 24-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N3583

☒ Gouge - Particle Size, Size Distribution: < 125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.1
Horz. Load Initial Value (kN): -0.5

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 5e6 μ m/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 3 - 10 - 30 - 100 - 300

Holds: 1 - 3 - 10 - 30 - 100 - 300 - 1000 - 3000

Temperature (°C): RT

Relative Humidity (%): dry

Comments:

thinnest square used for setup

605.57g 614.19g
604.86g 613.40g

HighSTEPS Experiment

Exp. Name: h418
Operator: Barnaby

Date: 25-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N3583

☒ Gouge - Particle Size, Size Distribution: 425 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.05

Horz. Load Initial Value (kN): -0.27

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 5e6 μ m/s²

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): wet

Comments:

thinnest square used for setup

saturated @ 1MPa 8:32-9:12

post mortem sample somewhat dry (~20%) and difficult to remove

604,83 613,19
605,78 613,77

HighSTEPS Experiment

Exp. Name: h366

Date: 1.9.22

Operator: Florian Bernold

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm

☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm

☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm

☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm

☐ Porous Frits: _____ mm

Spacers: thinner mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N7808

☒ Gouge - Particle Size, Size Distribution: < 125 μ m

☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0.05

Horz. Load Initial Value (kN): -0.45

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 11.56 (kN) 6 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s

Acceleration: 10⁶ (5-10⁶) μ m/s²

Deceleration: "

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 3 - 6 - 30 - 100 - 300

Holds: 1 - 3 - 10 - 30 - 100 - 300 - 1000 - 3000

Temperature (°C): RT

Relative Humidity (%): RH

Comments:

thinnest square for sample prep

605,48 613,67
604,85 613,54

HighSTEPS Experiment

Exp. Name: h367

Date: 1.9.22

Operator: Flora Bernady

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm

☐ Small Bare Surface (3.4 cm x 2.0 cm): mm

☐ Large Bare Surface (3.4 cm x 4.0 cm): mm

☐ Vessel Bare Surface (3.4 cm x 2 cm): mm

☐ Porous Frits: mm

Spacers: thinnest mm

Total Thickness-Bench: mm

Material (Quartz, Granite, ...?): N3808

☒ Gouge - Particle Size, Size Distribution: <125 μ

☐ Bare Surface - Roughness, Grit:

Vert. Load Initial Value (kN): -0.01

Horz. Load Initial Value (kN): -0.5

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.01

Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: (MPa)

Pore Pressure SX: (MPa)

Pore Pressure DX: (MPa)

Sliding Velocity: 10 μ m/s

Acceleration: 16^g (5-16^g) μ m/s²

Deceleration:

Starting Position:

Shear Displ. Sample:

Velocity Steps: 10-3-10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): RT saturated

Comments:

thinnest square for sample prep

saturation: 15604 - 16604

Mega suction, difficult to remove sample (and clean)

velocity steps a bit ugly, noise on FV on tuning

606,09 614,58
606,49 614,93

HighSTEPS Experiment

Exp. Name: h 769
Operator: Florian

Date: 7.9.22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 74x74 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: _____ mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N 7808

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0.1

Horz. Load Initial Value (kN): -0.71

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 74.68 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: _____

Acceleration: _____

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: _____

Holds: _____

Temperature (°C): RT

Relative Humidity (%): RH

Comments:

Heinrich square for sample prep

605.46g 613.26g
604.89g 613.27g

HighSTEPS Experiment

Exp. Name: N368
Operator: Florian

Date: 7-9-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N3808
☐ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0.32
Horz. Load Initial Value (kN): -0.06

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.002
Td: 0.001

Normal Stress(es): 34.68 (kN) 70 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/sec

Acceleration: 10^6 (5-10⁶) μ m/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 3 - 10 - 30 - 100 - 700

Holds: 1 - 3 - 10 - 30 - 100 - 700 - 1000 - 3000

Temperature (°C): RT
Relative Humidity (%): saturated

Comments:

Thinnest spacer used for setup

saturation: 86.27 - 91.07

start SHS at 1646

606.13g 614.28
606.55g 614.56g

HighSTEPS Experiment

Exp. Name: h419
Operator: Burnaby

Date: 25-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N3808

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.15
Horz. Load Initial Value (kN): -0.5

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 57.8 (kN) SD (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 16mm

Acceleration: 5e6 μ m/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300-1000
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): dry

Comments:

thinnest square used for setup

605.55g 613.69g
604.89g 612.44g

HighSTEPS Experiment

Exp. Name: h420
Operator: Barnaby

Date: 26-10-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N3808

☒ Gouge - Particle Size, Size Distribution: 425 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.07
Horz. Load Initial Value (kN): -0.38

Horizontal Load PID Setting:

gain: 0.008
Ti: 0.07
Td: 0.001

Normal Stress(es): 57.8 (kN) 9 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 566 μ m/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): wet

Comments:

thinnest square used for setup

Saturated @ 1MPa 9:22 - 10:03

Post mortem sample mostly wet, but some dry (~10%)

sample holders 605,55 615,17
604,86 617,59

HighSTEPS Experiment

Exp. Name: h 319

Date: 10.5.22

Operator: Bernady / Florian

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm

☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm

☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm

☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm

☐ Porous Frits: _____ mm

Spacers: thin one _____ mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N2 - D = 4330 Pt Newbury

☒ Gouge - Particle Size, Size Distribution: < 125 μ m

☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.1

Horz. Load Initial Value (kN): -0.36

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s

Acceleration: 10⁶ (5-10⁶) μ m/s²

Deceleration: 11

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 3 - 10 - 30 - 100 - 300

Holds: 1 - 3 - 10 - 30 - 100 - 300 - 1000 - 3000

Temperature (°C): RT

Relative Humidity (%): RH

Comments:

load point at 7.5
8.39

thinnest square used for sample prep

39.35
28.6
21.2

sample holder 605, 50 614, 92
604, 88 614, 39

HighSTEPS Experiment

Exp. Name: h 320
Operator: Barudy Florian

Date: 10.5.22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thin on _____ mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N2 - 4330
☒ Gouge - Particle Size, Size Distribution: N2 - 4330 < 125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.22
Horz. Load Initial Value (kN): -0.16

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 11.56 (kN) _____ (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s

Acceleration: 10⁶ (5.10⁶) μ m/s²
Deceleration: _____

Starting Position: _____
Shear Displ. Sample: _____

Velocity Steps: _____
Holds: _____

Temperature (°C): RT
Relative Humidity (%): wet

Comments:

thinnest square used for sample prep

touchpoint 7.3 -
welding @ 1402
1442

605.49g 614.55g
604.85g 613.74g

HighSTEPS Experiment

Exp. Name: h466
Operator: Barnaby

Date: 30-11-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N4330

☒ Gouge - Particle Size, Size Distribution: 425 μ m

☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.42

Horz. Load Initial Value (kN): -0.3

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 34.68 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 5e6 μ m/s²

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): Rt

Relative Humidity (%): dry

Comments:

thinnest square used for setup
last one (probably)
I hate myself :)

$P_I = 604.86g$ $P_{I\text{change}} = 613.91g$
 $P_{II} = 605.51g$ $P_{II\text{change}} = 614.24g$

HighSTEPS Experiment

Exp. Name: 11338
Operator: Florian - Arthur

Date: 25/07/2022

Sample Holders Thickness w/o Gouge or Bare Surface:

☐ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): Newberry N2 12-3 D=4330 ft

☒ Gouge - Particle Size, Size Distribution: 125 μm

☐ Bare Surface - Roughness, Grit: _____

Newberry N2

Vert. Load Initial Value (kN): -0.1

Horz. Load Initial Value (kN): -0.19

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 34680 (kN) 30 (MPa)

Confining Pressure: / (MPa)

Pore Pressure SX: / (MPa)

Pore Pressure DX: / (MPa)

Sliding Velocity: 10 $\mu m s^{-1}$

Acceleration: _____

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 3-10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000

Temperature ($^{\circ}C$): RT

Relative Humidity (%): Wet ()

Comments: thinnest square

wetting: 12h43 - 13h26

oscillation of about 0.2kN during wetting at normal force

606.14g 614.98g
606.56g 615.68g

HighSTEPS Experiment

Exp. Name: N439
Operator: Balch

Date: 8-11-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N4330

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.3
Horz. Load Initial Value (kN): -0.7

Horizontal Load PID Setting:

gain: 0.03
Ti: 0.02
Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 5e6 μ m/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 30 - 100 - 300

Holds: 1 - 3 - 10 - 30 - 100 - 300 - 1000 - 3000

Temperature (°C): RT

Relative Humidity (%): dry

Comments:

thinnest square used for setup

605.48g 614.34g
604.84g 613.77g

HighSTEPS Experiment

Exp. Name: h440
Operator: Barnaby

Date: 9-11-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N4320

☒ Gouge - Particle Size, Size Distribution: 425 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.25

Horz. Load Initial Value (kN): -0.4

Horizontal Load PID Setting:

gain: 0.05

Ti: 0.02

Td: 0.00

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 1mm

Acceleration: 5e6 μ m/s²

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): wet

Comments:

thinnest square used for xhp

Saturated @ 1 MPa 9:02-9:43

1000 second hold reset due to base oscillations
I then cancelled 3000 second thinking it was too long ;)

605.48

614.58

A

604.67

613.93

HighSTEPS Experiment

Exp. Name: h376Date: 15-9-22Operator: Florent Bernabé

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm☐ Porous Frits: _____ mmSpacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N4343☒ Gouge - Particle Size, Size Distribution: <125 μ m☐ Bare Surface - Roughness, Grit: _____Vert. Load Initial Value (kN): 0.1Horz. Load Initial Value (kN): -0.3

Horizontal Load PID Setting:

gain: 0.03Ti: 0.02Td: 0.02Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/sec for 11mmAcceleration: 5e6 μ m/s²

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-200Holds: 1-3-10-20-100-200-1000-2000Temperature (°C): RTRelative Humidity (%): RH

Comments:

Same sample as h375, consider to repeat as already slightly compacted once

thinnest square for sample prep

held 10 MPa 14:42 - 15:02

606.12g 614.57g
606.52g 615.15g

HighSTEPS Experiment

Exp. Name: h379
Operator: Barnaby

Date: 26-9-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N4343
☐ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0.15
Horz. Load Initial Value (kN): -0.22

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11 mm

Acceleration: 526 μ m/s²
Deceleration: _____

Starting Position: _____
Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300
Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT
Relative Humidity (%): wet

Comments:

thinnest square used for setup
saturated 10:14 - 10:54 @ 1 MPa
held 10 MPa 11:11 - 11:31

606.12g 614.89g
606.55g 615.21g

HighSTEPS Experiment

Exp. Name: h377
Operator: Barnaby

Date: 16-9-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N4343

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.15
Horz. Load Initial Value (kN): -0.625

Horizontal Load PID Setting:

gain: 0.063
Ti: 0.02
Td: 0.001

Normal Stress(es): 34.68 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/sec for 11mm

Acceleration: 5e6 μ m/sec²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): RH

Comments:

Thinnest square used for setup
held 30 MPa 10:15 - 10:34

Noise on 100 μ m/s velocity step

605.52g 614.18g
604.85g 613.54g

HighSTEPS Experiment

Exp. Name: H378
Operator: Barnaby

Date: 16-9-22
~12:45

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest _____ mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N4343

☒ Gouge - Particle Size, Size Distribution: < 125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.15
Horz. Load Initial Value (kN): -0.6

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 34.68 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/sec for 11mm

Acceleration: 5e6 μ m/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): Wet

Comments:

thinnest square used for setup

Saturation 13:00 - 13:41

Held 30 MPa 14:56 - 14:56

606.13g 614.87g
606.54g 615.56g

HighSTEPS Experiment

Exp. Name: h441
Operator: Barnaby

Date: 9-11-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N4343

☒ Gouge - Particle Size, Size Distribution: 425 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.32
Horz. Load Initial Value (kN): -0.45

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 500 μ m/s²
Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-200

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): dry

Comments:

thinnest square used for setup

606.19g 614.90g
606.56g 615.53g

HighSTEPS Experiment

Exp. Name: h463
Operator: Burnaby

Date: 29-11-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N4343

☒ Gouge - Particle Size, Size Distribution: 425 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.35

Horz. Load Initial Value (kN): -0.13

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 56 μ m/s²

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10-3-10-30-100-200

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): wet

Comments:

thinnest square used for setup

Saturated @ 1MPa 9:33-10:14

Small leak already during saturation period
but we powered through home

606.16 614,97
606.54 615,11

HighSTEPS Experiment

Exp. Name: 4377
Operator: T. Loner

Date: 9.9.22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 74x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N 4422

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.107

Horz. Load Initial Value (kN): -0.176

Horizontal Load PID Setting:

gain: 0.1003

Ti: 0.102

Td: 0.1001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s

Acceleration: 10⁶ (5.10⁶) μ m/s²

Deceleration: "

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 6-7-10-30-100-300

Holds: 1-3-10-30-100-300-1000-3000

Temperature (°C): RT

Relative Humidity (%): RT

Comments:

thinest square for sample prep

HighSTEPS Experiment

Exp. Name: 4772
Operator: _____

Date: 9.9.22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 77 mm
____ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
____ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
____ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
____ Porous Frits: _____ mm

Spacers: _____ mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N4422

☒ Gouge - Particle Size, Size Distribution: < 125 μ m
____ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): ~~20.76~~ 0.107

Horz. Load Initial Value (kN): -0.16

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 11.56 (kN) 10 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s

Acceleration: 10⁶ (5.6⁶) μ m/s²

Deceleration: "

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 3 - 10 - 20 - 100 - 300

Holds: 1 - 3 - 10 - 30 - 100 - 300 - 1000 - 3000

Temperature (°C): RT

Relative Humidity (%): saturated

Comments:

thinnest square for sample prep
saturation: 7h58 - 8h39
belle

HighSTEPS Experiment

Exp. Name: h371
Operator: Florian

Date: 8.9.22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34x34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N4422

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.02

Horz. Load Initial Value (kN): -0.33

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.02

Td: 0.001

Normal Stress(es): 34.68 (kN) 30 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s

Acceleration: 10⁶ (5.169) μ m/s²

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 3 - 10 - 30 - 100 - 300

Holds: 1 - 3 - 10 - 30 - 100 - 300 - 1000 - 3000

Temperature (°C): RT

Relative Humidity (%): RH

Comments:

thinnest square for sample prep

605,48 614,46
604,85 614,13

HighSTEPS Experiment

Exp. Name: h 370
Operator: Florian

Date: 8.9.22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 74x37 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N4422

☒ Gouge - Particle Size, Size Distribution: <125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): 0.16
Horz. Load Initial Value (kN): -0.24

Horizontal Load PID Setting:

gain: 0.003
Ti: 0.02
Td: 0.001

Normal Stress(es): 34.68 (kN) 70 (MPa)

Confining Pressure: _____ (MPa)
Pore Pressure SX: _____ (MPa)
Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s

Acceleration: 10⁶ (5-10⁶) μ m/s²
Deceleration: "

Starting Position: _____
Shear Displ. Sample: _____

Velocity Steps: 10 - 3 - 10 - 70 - 100 - 700
Holds: 1 - 3 - 10 - 70 - 100 - 700 - 1000 - 3000

Temperature (°C): RT
Relative Humidity (%): saturated

Comments:

thinnest square for sample preparation
saturation: 74.59 - 84.40

606.12g 614.53g
606.54g 614.89g

HighSTEPS Experiment

Exp. Name: h443
Operator: Barnaby

Date: 10-11-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N4422

☒ Gouge - Particle Size, Size Distribution: ~125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.3

Horz. Load Initial Value (kN): -0.52

Horizontal Load PID Setting:

gain: 0.003

Ti: 0.52

Td: 0.001

Normal Stress(es): 57.8 (kN) 50 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 50 μ m/s²

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 - 3 - 10 - 30 - 100 - 200 - 1000 - 2000

Holds: 1 - 3 - 10 - 30 - 100 - 200 - 1000 - 2000

Temperature (°C): RT

Relative Humidity (%): dry

Comments:

thinnest square used for setup

605.54g 614.33g
604.91g 613.20g

HighSTEPS Experiment

Exp. Name: N444
Operator: Bernie

Date: 11-11-22

Sample Holders Thickness w/o Gouge or Bare Surface:

☒ Simple Grooves (3.4 cm x 3.4 cm): 34 x 34 mm
☐ Small Bare Surface (3.4 cm x 2.0 cm): _____ mm
☐ Large Bare Surface (3.4 cm x 4.0 cm): _____ mm
☐ Vessel Bare Surface (3.4 cm x 2 cm): _____ mm
☐ Porous Frits: _____ mm

Spacers: thinnest mm

Total Thickness-Bench: _____ mm

Material (Quartz, Granite, ...?): N4422

☒ Gouge - Particle Size, Size Distribution: 2125 μ m
☐ Bare Surface - Roughness, Grit: _____

Vert. Load Initial Value (kN): -0.2

Horz. Load Initial Value (kN): -0.35

Horizontal Load PID Setting:

gain: 0.03

Ti: 0.02

Td: 0.001

Normal Stress(es): 57.8 (kN) 80 (MPa)

Confining Pressure: _____ (MPa)

Pore Pressure SX: _____ (MPa)

Pore Pressure DX: _____ (MPa)

Sliding Velocity: 10 μ m/s for 11mm

Acceleration: 526 μ m/s²

Deceleration: _____

Starting Position: _____

Shear Displ. Sample: _____

Velocity Steps: 10 μ m/s - 10 μ m/s - 100 μ m/s

Holds: 1 μ m/s - 10 μ m/s - 100 μ m/s - 1000 μ m/s

Temperature (°C): RT

Relative Humidity (%): wet

Comments:

thinnest gauge used for setup
saturated @ 1 MPa 9:08 - 9:48